

Suiyang Guo

+86 158-9334-6463 | 1327217178@qq.com | Northwestern Polytechnical Univ. | MSc,
Control Eng. | 2024.09—2027.06



Control Engineering MSc candidate at Northwestern Polytechnical University (Class of 2027), targeting embedded software and robotic systems roles. Proficient in C/C++, STM32, FreeRTOS, embedded Linux and ROS/ROS2, with hands-on experience across firmware, motor control, PCB design, host applications and full-system integration. Built a Yocto/i.MX6ULL robot controller, in-house dual-arm/chassis motor-control systems and real-time VLA deployment pipelines. Contributor to a national key robotics project; holder of 6 patents and 2 software copyrights.

Education

Northwestern Polytechnical University (NPU)	2024.09—2027.06
MSc, Control Engineering Robotics & AI GPA: 89.56/100 (Top 30%)	
Zhengzhou University	2020.09—2024.06
BEng, Automation (Elite Class) Robotics & AI GPA: 3.54/4.0 (Top 20%)	

Technical Skills

- **Programming & Systems:** C/C++, Python and MATLAB; Linux driver/application development, Git, debugging and multi-module system integration
- **Embedded Linux / RTOS:** Yocto, i.MX6ULL, STM32F4, 8051, Arduino and FreeRTOS; board bring-up, lower-level control, sensor interfaces and motor drivers
- **Robotics:** ROS/ROS2, navigation, SLAM, motion planning, inverse kinematics, point-cloud processing, collision detection and fuzzy-PID control
- **Hardware & Applications:** Altium Designer/JLCPCB, peripheral circuits, Raspberry Pi, PyQt5 host applications and 3D printing
- **Edge AI & Performance:** PyTorch/JAX deployment, Triton kernels, CUDA Graphs and RTC; experience integrating AI policies into real-time robot systems

Embedded & Robotics Projects

Series-Parallel Hybrid Wheeled Humanoid Robot	2024.09—Present
National Key Research Project Key Provincial Lab, Team Member	
• Integrated a wheeled humanoid comprising an AGV base, parallel mechanism, dual 6-DoF arms and a 2-DoF head • Developed the C++/ROS Noetic master controller for 3D mapping, path planning, object recognition, dynamic obstacle avoidance, point-cloud publishing, collision detection and inverse kinematics • Built a MuJoCo simulation/control-integration environment and developed in-house dual-arm and chassis motor-control systems for actuator and full-robot bring-up • Built a Yocto-based embedded-Linux image for i.MX6ULL, ran roscore and chassis control on-board, and implemented wired-Ethernet ROS multi-machine communication • Developed STM32F4 lower-level position/chassis control and optimized the multi-DoF arm-control scheme, doubling the 6-DoF arm control frequency	
Three Generations of TCM Therapy Devices	2021.03—2023.09
Challenge Cup Provincial Second Prize Core R&D Member	
• Led electronics and software development across three product generations, spanning circuits/PCB, embedded firmware, host applications and system integration • Built STM32/FreeRTOS control systems with fuzzy PID, temperature/image/pressure sensing, Bluetooth communication and multi-task scheduling • Designed STM32 peripheral circuits and PCBs; developed servo/stepper drivers and a smoke-handling module; supported iterative prototype bring-up and debugging	

- Independently built a Raspberry Pi 4B/Python/PyQt5/ROS voice robot and host application
- Integrated audio capture, ASR, LLM dialogue, speech synthesis, digital-human lip sync and smart-home control; received 2 software copyrights

Engineering Experience

IDEA Research: Visincept

2026.04—Present

- VLA & World Model Intern | Robot Systems and Edge AI
- Deployed dual-arm VLA policies and migrated the inference service from ROS1 to ROS2, implementing a dual-clock control pipeline with 30 Hz command publishing and 3 Hz model inference
 - Applied RTC, CUDA Graphs and Triton autotuning to reduce step latency from 256.5 ms to 32.05 ms (8×) and idle end-effector jitter by 81%
 - Brought up X-VLA on the in-house dual-arm platform and used firmware EndPoseCtrl IK to reduce the control cycle from 1.4 s to 0.33 s
 - Built preflight safety checks, continuous-gripper mapping and WebSocket/ROS2 interfaces for model-to-robot integration

Zhongqin Herui (Shaanxi) Technology Co., Ltd.

2025.11—2026.01

- Technical Partner, Part-time
- Selected and calibrated sensors and developed 3D mapping/navigation for an autonomous orchard inspection vehicle
 - Built 3D maps from raw LiDAR and converted them into occupancy grids for navigation

Awards & Intellectual Property

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| • Challenge Cup Henan Collegiate Academic Sci-Tech Works Competition | Provincial Second Prize (2023) |
| • Collegiate Electrical & Electronic Engineer Innovation Contest | Central China First Prize (2023) |
| • Siemens Cup Intelligent Manufacturing Challenge | Western Region Second Prize (2022) |
| • Lanqiao Cup, Embedded Design & Development | Provincial Third Prize (2022) |
| • 4 invention patents, 2 utility-model patents and 2 software copyrights; CET-6: 473 | |

